

Celebal Summer Internship 2026 – SQL Week 3

Mini Project: Customer Sales Insights

Internship: Celebal Summer Internship 2026 – SQL Week 3

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Objective

The objective of this mini project is to analyze the Superstore sales dataset using advanced SQL concepts such as Subqueries, Common Table Expressions (CTEs), JOINS, and Window Functions. The analysis focuses on identifying customer purchasing patterns, ranking customers based on sales, and generating business insights to support data-driven decision-making.

1. Top 5 Customers

SQL Query

```
query = """
SELECT
    customer_name,
    SUM(sales) AS TotalSales
FROM orders o
JOIN customers c
ON o.customer_id = c.customer_id
GROUP BY customer_name
ORDER BY TotalSales DESC
LIMIT 5;
"""

pd.read_sql(query, conn)
```

	customer_name	TotalSales
0	Sean Miller	25043.050
1	Tamara Chand	19052.218
2	Raymond Buch	15117.339
3	Tom Ashbrook	14595.620
4	Adrian Barton	14473.571

Insight

The top five customers generated the highest total sales among all customers. These customers contribute significantly to overall business revenue and should be prioritized through loyalty programs, personalized offers, and premium customer engagement strategies.

2. Bottom 5 Customers

SQL Query

```
query = """
SELECT customer_name, SUM(sales) AS TotalSales
FROM orders o
JOIN customers c
ON o.customer_id = c.customer_id
GROUP BY customer_name
ORDER BY TotalSales ASC
LIMIT 5;
"""

pd.read_sql(query, conn)
```

	customer_name	TotalSales
0	Thais Sissman	4.833
1	Lela Donovan	5.304
2	Carl Jackson	16.520
3	Mitch Gastineau	16.739
4	Roy Skaria	22.328

Insight

The bottom five customers contributed the least amount of sales. Their purchasing activity is relatively low, indicating opportunities for targeted promotions, customer retention campaigns, or personalized marketing to improve engagement.

3. Customers Who Made Only One Order

SQL Query

```
query = """
SELECT customer_name, COUNT(DISTINCT order_id) AS TotalOrders
FROM orders o JOIN customers c
ON o.customer_id = c.customer_id
GROUP BY customer_name
HAVING COUNT(DISTINCT order_id) = 1;
"""

pd.read_sql(query, conn)
```

	customer_name	TotalOrders
0	Anemone Ratner	1
1	Anthony O'Donnell	1
2	Carl Jackson	1
3	Jenna Caffey	1
4	Jocasta Rupert	1
5	Lela Donovan	1
6	Mitch Gastineau	1
7	Patricia Hirasaki	1
8	Ricardo Emerson	1
9	Roland Murray	1
10	Susan MacKendrick	1
11	Theresa Coyne	1

Insight

These customers placed only one order during the recorded period. Identifying one-time buyers helps the business understand customer retention challenges and design strategies to encourage repeat purchases.

4. Customers with Above-Average Sales

SQL Query

```
query = """
WITH CustomerSales AS
(SELECT customer_id, SUM(sales) AS TotalSales FROM orders GROUP BY customer_id
)
SELECT c.customer_name, cs.TotalSales
FROM CustomerSales cs
JOIN customers c
ON cs.customer_id = c.customer_id
WHERE cs.TotalSales >
( SELECT AVG(TotalSales) FROM CustomerSales
);
"""

pd.read_sql(query, conn)
```

	customer_name	TotalSales
0	Brosina Hoffman	6255.3510
1	Irene Maddox	4930.4740
2	Pete Kriz	8646.9340
3	Tracy Blumstein	4737.4860
4	Matt Abelman	4299.1610
...	...	...
289	Sarah Brown	6410.9960
290	Trudy Schmidt	3368.0940
291	Victoria Pisteka	3360.5260
292	Craig Molinari	3984.4524
293	Lindsay Castell	3246.6260

294 rows x 2 columns

Insight

Customers whose total sales exceed the overall average represent the business's high-value customer segment. Maintaining strong relationships with these customers can improve long-term revenue and customer lifetime value.

5. Highest Order Value per Customer

SQL Query

```
query = """
SELECT customer_name, MAX(sales) AS HighestOrder
FROM orders o
JOIN customers c
ON o.customer_id = c.customer_id
GROUP BY customer_name;
"""

pd.read_sql(query, conn)
```

	customer_name	HighestOrder
0	Aaron Bergman	341.960
1	Aaron Hawkins	668.160
2	Aaron Smayling	1439.982
3	Adam Bellavance	4355.168
4	Adam Hart	841.568
...	...	...
788	Xylona Preis	337.088
789	Yana Sorensen	2793.528
790	Yoseph Carroll	2934.330
791	Zuschuss Carroll	1516.200
792	Zuschuss Deastell	833.060

Insight

This analysis identifies the highest-value order placed by each customer. It provides insights into customers' maximum spending potential and helps understand purchasing behavior across different customer segments.

Overall Findings

- A few customers generate most of the sales.
- Customer spending is different from one customer to another.
- Some customers placed only one order, so they may need better engagement.
- High-value customers should be rewarded with loyalty programs.
- SQL features like **JOINS**, **CTEs**, **Subqueries**, and **Window Functions** help analyze customer data easily

Conclusion

The Customer Sales Insights analysis demonstrates how advanced SQL techniques can be used to transform raw sales data into meaningful business intelligence. By identifying customer rankings, purchasing behavior, and sales trends, organizations can make informed decisions to improve customer retention, optimize marketing strategies, and increase overall business performance. This mini project showcases the practical application of SQL for solving real-world business analytics problems.